IoT Project

Weather Station with focus on Rain Detection and user alerting when rain stops

Josh Jameson - S00291620 Ahmad Din - S00287223 Arif Taha - S00292948 Paschal Okolo - S00292938

Problems

1. Hard to find libraries in C#
2. Pi has no analogue pins hence extra chip needed to convert reading

Addressing threats

- Data encryption (HTTP/HTTPS)
- Creation of anomaly detecting code by adding reading validators

Methods of execution

- Deployment of C# to build the weather station code
- Use of Dupont wires to connect the sensors to the breadboard and Raspberry Pi
- rain, temperature and humidity sensors to collect our readings

SDG's

- Life on Land
- Climate Action

Our Idea

We strove to make a fully functional miniature weather station with the code based off of C#.
This weather station would have the ability to sense rainfall, temperature & humidity as well as the time of day (both day & night and precise time telling).

Main Features

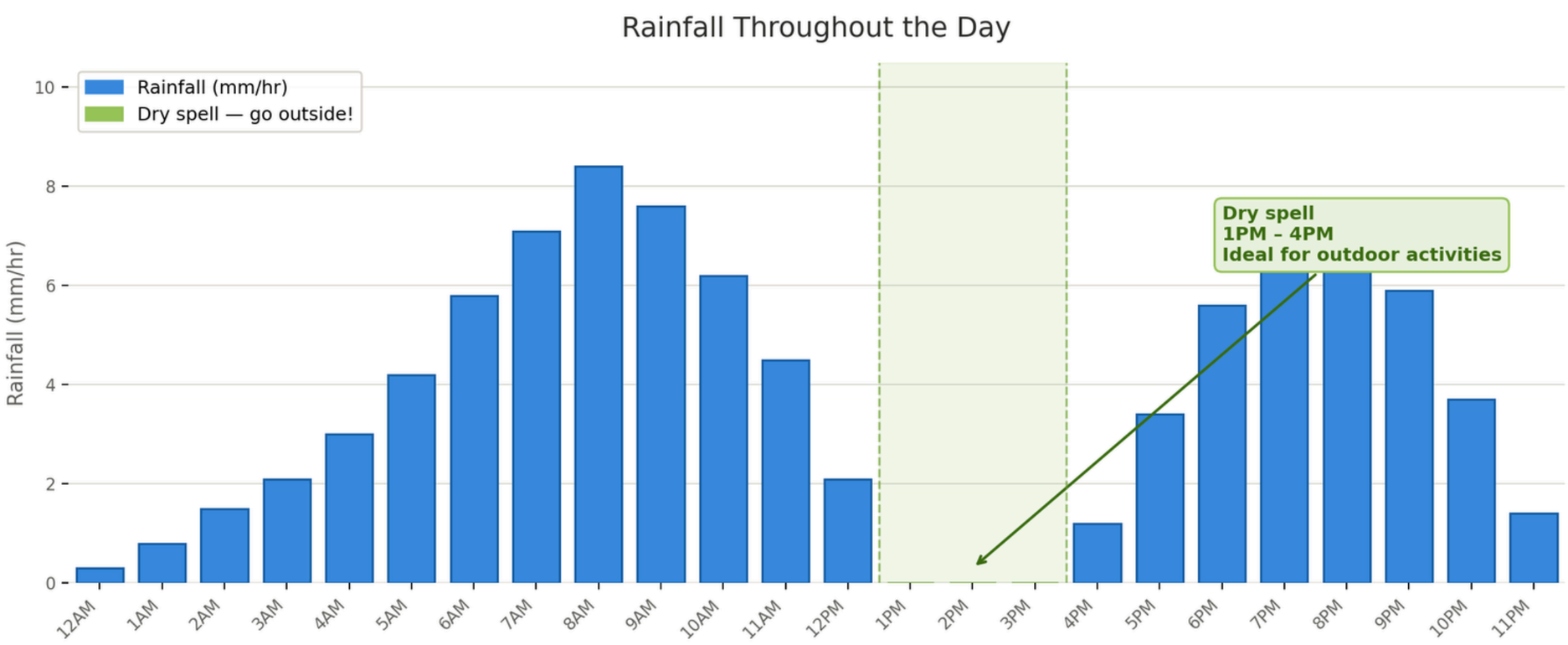
- Rain detection
- Humidity detection
- Temperature detection
- Direct alerts to smart phone

Communication

1. Whatsapp
2. Email
3. Scheduled Meetings
4. GitHub

Security threats

1. Leaking of private end user info
2. Data fabrication
3. Illegal selling of data



Tools Used

- GitHub
- Visual Studio
- Canva
- Figma
- Pushing Box